Assignment 9 - Entanglement Swapping

1. Problem statement

Consider the quantum network shown in Figure 1. The network consists of two *quantum end* nodes, Alice and Bob, and N *quantum repeater* nodes located between them.

The network supports only the generation of entangled Bell pairs between *adjacent* nodes. The end nodes are assumed to have limited quantum hardware: each can store only one qubit in quantum memory and perform single-qubit operations. The repeater nodes are more capable quantum processors: each repeater can store two qubits simultaneously and apply two-qubit operations, including *Bell-state measurements* (BSMs) on the stored qubits.

The objective is to establish *end-to-end entanglement* between Alice and Bob by means of the *entanglement swapping* protocol executed at the repeater nodes.

Figure 1 illustrates:

1. the general topology with N repeaters,
2. an example instance with $N = 3$, and
3. the final end-to-end Bell pair shared by Alice and Bob.

2. Ideal benchmark

The network should first be evaluated in an *ideal setting*, in order to verify the correctness of the implementation and the expected behavior of the entanglement swapping protocol.

In this ideal case:

- Bell-pair generation on each adjacent link is assumed to be perfect and instantaneous.
- Quantum memories are noiseless.
- Bell-state measurements are ideal.

3. Non-ideal model

After validating the ideal case, the following realistic assumptions should be introduced.

- *Memory noise*: Each quantum memory slot is affected by *depolarizing* noise with parameter p_w. For simplicity, it is acceptable to assume that all memory locations, both at the end nodes and at the repeater nodes, are subject to the same depolarizing model.

- *Probabilistic Bell-pair generation*: Bell pairs generated between adjacent nodes are assumed to be perfect, but their preparation is *probabilistic*. A single attempt to generate a Bell pair can be modeled as a Bernoulli experiment with success probability p_{success} and failure probability $1 - p_{\text{success}}$. Each attempt takes one discrete time unit. Therefore, the number of attempts required to successfully generate a Bell pair follows a *geometric* distribution.

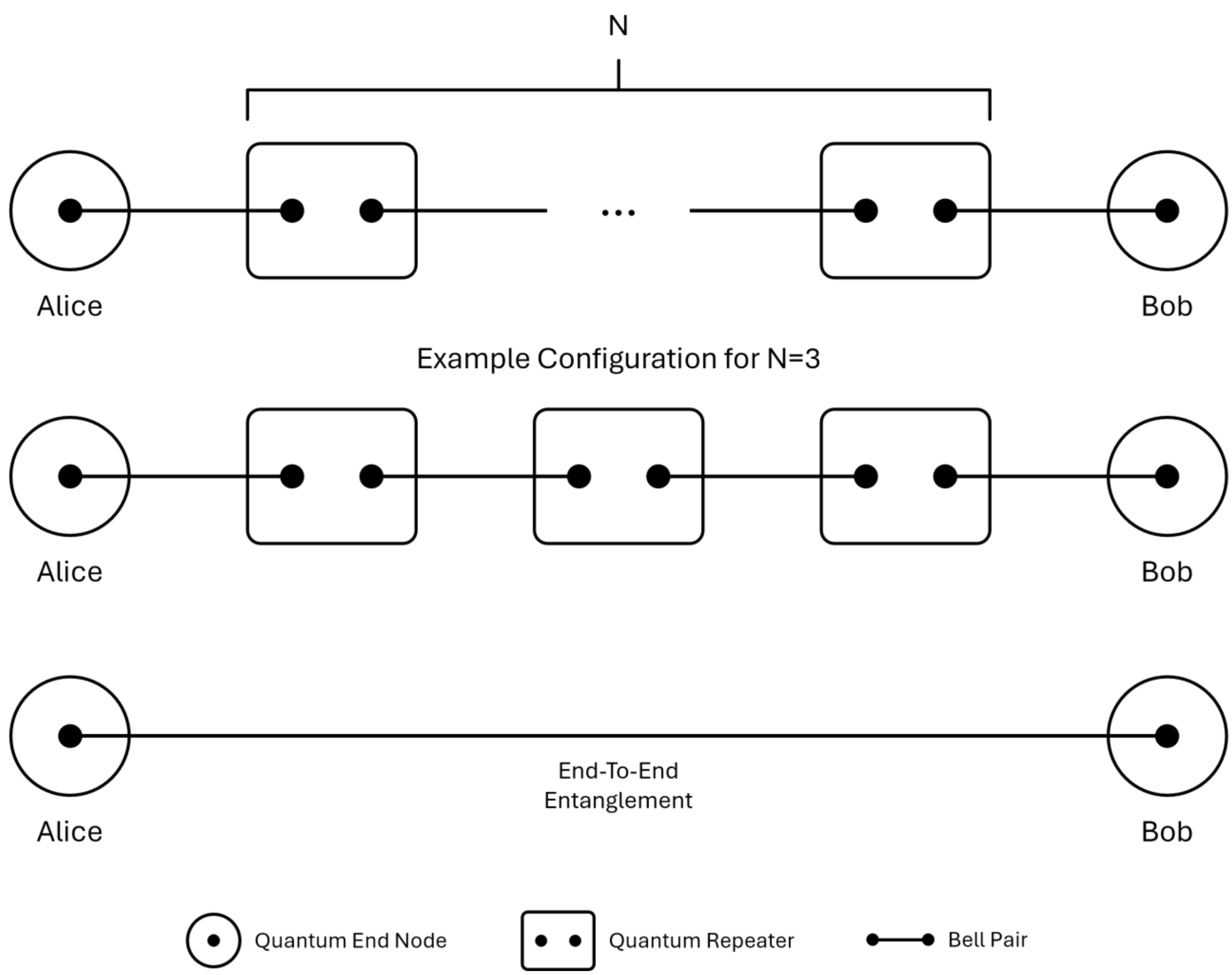

Figure 1: Quantum network topology for entanglement swapping: general chain of N repeater nodes between Alice and Bob, an example configuration with N = 3, and the resulting end-to-end entangled Bell pair.

4. Performance metrics

The following quantities should be measured and analyzed:

- *Fidelity of the final entangled state*: The final fidelity of the end-to-end Bell pair shared by Alice and Bob should be evaluated with respect to the ideal Bell state $|\Phi^+\rangle$:

$$F = \langle\Phi^+|\, \rho_{AB}\, |\Phi^+\rangle$$

  where ρ_{AB} is the final two-qubit state held by Alice and Bob.

- *Entanglement distribution time*: The entanglement distribution time is the number of time units required to establish the final end-to-end entangled state between Alice and Bob.

5. Required analysis and plots

The simulation study should investigate how the above parameters affect the performance metrics, with emphasis on the following plots:

- *Distribution time versus success probability*: Plot the entanglement distribution time as a function of p_{success}, with separate curves representing different values of N. Three curves corresponding to distinct values of N are sufficient. This plot should highlight how the entanglement generation time decreases as p_{success} increases, and how larger repeater chains affect latency.

- *Fidelity versus success probability*: Plot the final fidelity as a function of p_{success}, with separate curves representing different values of p_w. Three curves for distinct values of p_w are sufficient. This plot should show how memory decoherence degrades the final fidelity and how this effect interacts with the stochastic entanglement-generation process.

6. Expected discussion

The results should include a concise interpretation of the observed trends, together with theoretical expectations where applicable. In particular, the analysis should address how the entanglement distribution time depends on p_{success}, how it scales with the number of repeaters N, and how the final-state fidelity degrades as the memory depolarization parameter p_w increases. Any additional relevant effects emerging from the simulations should also be discussed. All observed trends should be clearly explained and, whenever possible, supported by theoretical arguments or validated against expected behavior.

7. Implementation requirements

The full quantum-network simulation and analysis must be carried out in *Julia*, using *QuantumSavory.jl* as the quantum network simulation framework.

8. Deliverables

The project deliverables are:

1. *Julia source code* implementing the simulation, analysis, and plot generation.
2. *Presentation slides* providing a general overview of the project, followed by a description of the adopted simulation workflow, the main results, and the final conclusions. The presentation should be designed to last no more than 15 minutes.